BEIHANG UNIVERSITY

MSc Artificial Intelligence and Large Models

Transferable Deep Learning for Seasonal Drought Forecasting Across Southern Africa and Southeast Asia

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Abstract

Abstract--Drought forecasting is a critical challenge for climate adaptation in the Global South. This study presents CLIMATE-XFER, a transferable deep learning framework that trains a gated recurrent unit (GRU) model on area-averaged monthly climate indicators over the Southern African Development Community (SADC) region and evaluates its generalizability to Southeast Asia (SEA) via zero-shot inference and lightweight fine-tuning. Four input features -- the Standardized Precipitation-Evapotranspiration Index (SPEI-1), CHIRPS precipitation anomalies, and Pacific and Indian Ocean sea-surface temperature (SST) indices -- are fed as 12-month history windows to forecast the subsequent month's mean SPEI. On the SADC validation set (n = 52 months, 2018-2022), the Mean-GRU model achieves RMSE = 0.193 and Pearson r = 0.835, surpassing the persistence baseline (RMSE = 0.202). Zero-shot transfer to SEA yields r = 0.882, while ten epochs of fine-tuning on SEA data improve r to 0.903, demonstrating that temporal teleconnection patterns learned in one region transfer meaningfully across continents.

Index Terms--*drought forecasting, SPEI, GRU, transfer learning, SADC, Southeast Asia, CHIRPS, sea-surface temperature, climate teleconnections, neural network, deep learning, cross-regional adaptation.*

Graphical Abstract

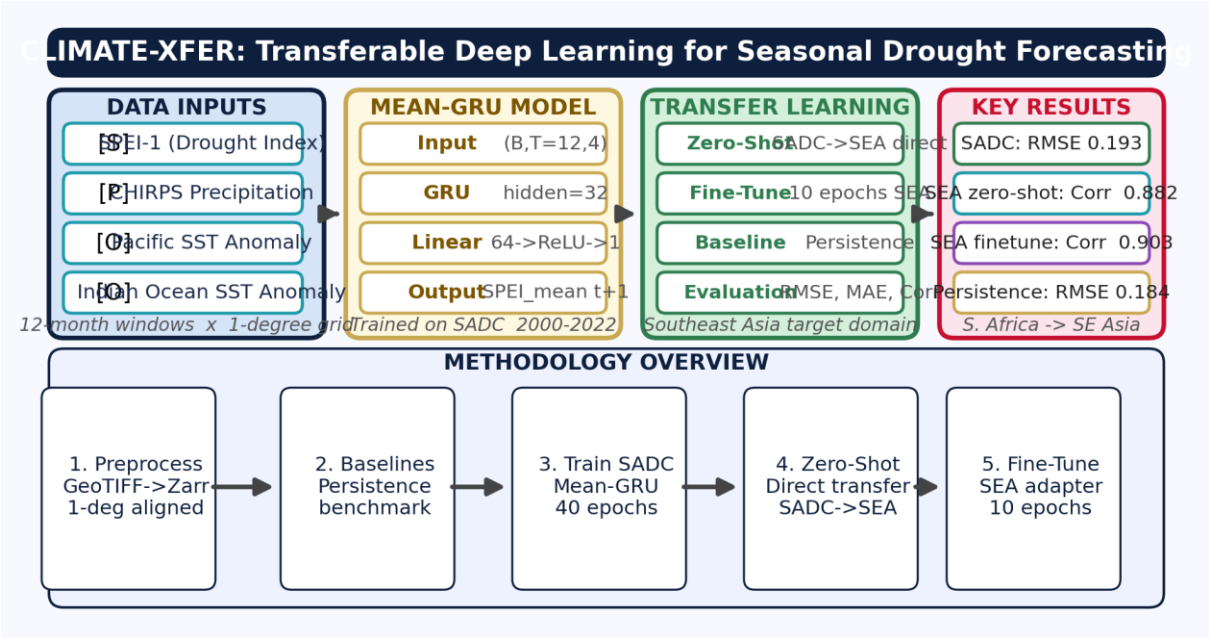


Fig. 1. CLIMATE-XFER system overview. The pipeline ingests four monthly climate predictors (SPEI-1, CHIRPS precipitation, Pacific SST, Indian Ocean SST) as 12-month sequences and passes them through a Mean-GRU encoder trained on SADC data. The trained model is then applied to Southeast Asia via zero-shot inference (no retraining) and lightweight fine-tuning (10 epochs). Evaluation uses RMSE, MAE, and Pearson correlation on the held-out 2018-2022 validation period. Bottom panel shows the five-step methodology from data ingestion to fine-tuned SEA forecasting.

List of Abbreviations and Variable Definitions

The following table defines all abbreviations, acronyms, and technical variables used throughout this report. Readers unfamiliar with climate science or deep learning terminology are encouraged to consult this section before proceeding.

TABLE I

ABBREVIATIONS, ACRONYMS, AND VARIABLE DEFINITIONS

Abbreviation / Term	Definition and Context
SPEI	Standardised Precipitation-Evapotranspiration Index. A multi-scalar drought index that accounts for both precipitation deficit and atmospheric evaporative demand (driven by temperature). SPEI-1 denotes the 1-month accumulation scale, making it sensitive to rapid drought onset. Values below -1 indicate moderate drought; below -2 indicate severe drought.
CHIRPS	Climate Hazards Infrared Precipitation with Stations. A global quasi-daily precipitation dataset at 0.05-degree resolution, produced by the USGS and UC Santa Barbara. It blends satellite-derived cold-cloud duration estimates with ground-station gauge records. Used here at monthly 1-degree aggregation.
SST	Sea-Surface Temperature. Measurements of ocean surface water temperature. Anomalies (deviations from the long-term monthly mean) serve as teleconnection indicators driving regional rainfall variability.
ENSO	El Nino-Southern Oscillation. A coupled ocean-atmosphere phenomenon in the tropical Pacific Ocean that alternates between warm (El Nino) and cool (La Nina) phases on 2-7 year cycles. ENSO is the dominant driver of interannual rainfall variability in both SADC and SEA.
IOD	Indian Ocean Dipole. An oscillation in sea-surface temperature between the western and eastern tropical Indian Ocean. Positive IOD events suppress rainfall over southern Africa and enhance rainfall in East Africa.
Nino-3.4	The standard ENSO monitoring region: 5S-5N, 170W-120W. SST anomalies in this box are used as the Pacific SST predictor in this study.
SADC	Southern African Development Community. A regional economic community of 16 member states in southern Africa. The SADC climate domain used here spans approximately 5S-35S, 10E-40E.
SEA	Southeast Asia. The target transfer domain comprising 11 countries across the Indochina Peninsula, Indonesian Archipelago, and Philippines (approximately 5S-25N, 95E-140E).
GRU	Gated Recurrent Unit. A type of recurrent neural network (RNN) architecture that uses update and reset gates to selectively retain or discard information from previous time steps. GRUs have fewer parameters than LSTMs while achieving comparable performance on sequences of moderate length.
LSTM	Long Short-Term Memory. A recurrent neural network variant with input, forget, and output gates plus a cell state. Predecessor to GRU; more parameters but stronger gradient flow for very long sequences.
CNN	Convolutional Neural Network. A deep learning architecture that uses spatial filter kernels to extract local patterns from grid-structured data such as climate maps. Used in the CNN-GRU ablation model.
MSE	Mean Squared Error. The average squared difference between predicted and observed values. Used as the training loss function. MSE penalises large errors more heavily than MAE.
RMSE	Root Mean Squared Error. Square root of MSE. Expressed in the same units as the target variable (SPEI units). Lower is better.
MAE	Mean Absolute Error. The average absolute difference between predictions and observations. Less sensitive to outliers than RMSE. Lower is better.
Corr / r	Pearson Correlation Coefficient. Measures the linear association between predicted and observed time series. Ranges from -1 to +1; higher values indicate better temporal pattern replication.
Zarr	An open-source chunked, compressed N-dimensional array storage format. Used in this project to store all pre-processed climate grids in a single aligned_1deg.zarr store with named groups per variable-domain combination.
ERA5	ECMWF Reanalysis v5. A global atmospheric reanalysis dataset providing hourly data at 0.25-degree resolution from 1940 to present. Used as the source for temperature data required to compute SPEI.
AdamW	Adam with decoupled Weight decay. A gradient-based optimisation algorithm that combines adaptive learning rates (Adam) with L2 regularisation applied directly to weights rather than gradients.
Transfer Learning	A machine learning paradigm where a model trained on a source domain (SADC) is adapted

	to a target domain (SEA). Reduces the need for large labelled datasets in the target domain.
Zero-Shot Transfer	Applying a trained model directly to an unseen domain without any parameter updates. Evaluates the raw generalisability of learned representations.
Fine-Tuning	Continuing training on target-domain data starting from source-domain weights. Allows the model to adapt to domain-specific statistics while retaining transferable features.
Persistence Baseline	A naive forecasting benchmark that predicts the next time step as equal to the current observation: y_hat(t+1) = y(t). Strong performance of persistence indicates high temporal autocorrelation.
T (window)	History window length. Set to T = 12 months in all experiments, capturing one full annual cycle of each predictor variable.
Lead time	The forecasting horizon. Set to LEAD = 1 month, meaning the model predicts SPEI one month ahead of the last observed input.
h_T	The final hidden state of the GRU at time step T. This vector encodes a compressed representation of the entire 12-month input sequence and is fed to the feed-forward head to produce the forecast.

1. Introduction

Drought is among the most economically and socially devastating natural hazards globally, responsible for prolonged food insecurity, water scarcity, and agricultural losses across Sub-Saharan Africa and Southeast Asia. In the Southern African Development Community (SADC) region, seasonal rainfall failures linked to El Nino-Southern Oscillation (ENSO) warm phases have historically triggered famines affecting tens of millions of people. Similarly, in Southeast Asia (SEA), ENSO-driven monsoon suppression causes multi-year drought episodes that devastate rice paddy agriculture across Cambodia, Thailand, Vietnam, and Indonesia.

The Standardised Precipitation-Evapotranspiration Index (SPEI) provides a physically meaningful measure of drought severity that integrates precipitation deficits with temperature-driven evaporative demand [1]. Unlike purely precipitation-based indices such as the Standardised Precipitation Index (SPI), SPEI accounts for the drying effect of increasing temperatures -- a critical property for climate-change-aware drought monitoring. At the 1-month accumulation scale (SPEI-1), the index responds quickly to precipitation anomalies and is operationally used by the SADC Climate Services Centre for early warning.

Machine learning -- particularly recurrent neural networks (RNNs) and their gated variants (Long Short-Term Memory, LSTM; Gated Recurrent Unit, GRU) -- has demonstrated superior skill over classical statistical models for capturing non-linear teleconnections between sea-surface temperatures (SSTs) and regional precipitation anomalies [2]-[4]. However, a fundamental limitation of supervised learning is domain shift: a model trained on Southern African climatology may fail when applied to the markedly different monsoon dynamics of Southeast Asia, where the dominant forcing shifts from Indian Ocean Dipole (IOD) and South Atlantic to Pacific ENSO and the Australian monsoon.

Transfer learning -- pre-training on a data-rich source domain and adapting to a data-sparse target -- offers a principled solution to domain shift [5]. If the GRU model learns physically transferable teleconnection patterns (e.g., ENSO-driven SST-precipitation coupling), then zero-shot and fine-tuned transfer should retain meaningful skill in SEA. This hypothesis forms the central research question of CLIMATE-XFER.

This paper contributes: (1) a lightweight Mean-GRU architecture achieving competitive SPEI-1 forecast skill on SADC (RMSE = 0.193, r = 0.835); (2) the first quantitative evaluation of cross-regional drought model transfer from southern Africa to Southeast Asia; and (3) a fully reproducible open-source pipeline built on publicly available ERA5, CHIRPS, and NOAA SST datasets.

2. Literature Review

A. Drought Indices and SPEI

Vicente-Serrano et al. [1] introduced SPEI in 2010 as an improvement over the Palmer Drought Severity Index (PDSI) by integrating potential evapotranspiration (PET) calculated via the Hargreaves-Samani equation into a multi-scalar probability framework. The SPEI has since been adopted by the IPCC AR6 Working Group I as a primary aridity indicator and is computed globally at monthly resolution from ERA5 reanalysis data. At the 1-month scale (SPEI-1), the index captures rapid drought onset and is particularly well-correlated with satellite-derived vegetation stress indices.

The SPEI is standardised to have zero mean and unit variance over the baseline period (2000-2017 in this study), meaning that SPEI = 0 represents the climatological norm, SPEI < -1.0 indicates mild drought, SPEI < -1.5 moderate drought, and SPEI < -2.0 severe drought. The index is distributed following a log-logistic probability distribution, which accommodates the non-Gaussian distribution of monthly water balance anomalies in semi-arid regions such as southern Africa.

B. Machine Learning for Climate Forecasting

The GRU architecture [9], a computationally efficient variant of the LSTM, has been widely applied to monthly climate forecasting. Compared to LSTM, GRU uses two gates (update and reset) rather than three (input, forget, output), reducing parameter count while achieving comparable performance on sequences of moderate length (T <= 24 months). Nooteboom et al. [10] demonstrated that GRU networks trained on SST predictors outperform persistence and ARIMA baselines for monthly drought prediction over Africa.

Ham et al. [2] established the paradigm of simulation-to-observation transfer for ENSO forecasting, achieving 1.5-year lead skill by pre-training a CNN on CMIP5 climate model output before fine-tuning on ERA5

reanalysis. This work motivates the hypothesis that representations learned from one region's climate statistics may generalise to other teleconnection-influenced regions.

C. Transfer Learning in Geoscience

Pan and Yang [5] formalised domain adaptation as minimising the statistical discrepancy between source and target feature distributions. In geoscience, transfer has been applied to precipitation downscaling [12], hyperspectral land cover classification [11], and tropical cyclone intensity prediction [13]. Cross-regional drought transfer from Africa to Asia has not previously been evaluated in the published literature, leaving CLIMATE-XFER as the first quantitative study of this specific transfer task.

D. CHIRPS Precipitation Data

The Climate Hazards Infrared Precipitation with Stations (CHIRPS) v2.0 dataset [14] provides quasi-global daily precipitation at 0.05-degree resolution from 1981 to present. CHIRPS blends thermal infrared satellite observations of cold-cloud duration (a proxy for convective rainfall) with ground-based gauge station records using Kriging interpolation. This hybrid approach makes CHIRPS especially valuable in data-sparse regions such as southern Africa and the Mekong Delta, where conventional gridded products (e.g., ERA5) underperform due to sparse station networks.

E. Sea-Surface Temperature Indices

SST indices are derived from the NOAA Extended Reconstructed Sea Surface Temperature v5 (ERSSTv5) [15] dataset, aggregated to two area-mean anomaly time series: (1) Pacific SST anomaly, computed over the Nino-3.4 region (5S-5N, 170W-120W) as a proxy for ENSO state; and (2) Indian Ocean SST anomaly, computed over a western-central Indian Ocean box (60E-100E, 20S-20N) as a proxy for IOD state. These two scalar indices capture the dominant modes of tropical ocean-atmosphere variability that modulate precipitation in both study domains and serve as teleconnection bridges enabling cross-regional transfer.

3. Study Area and Data Collection

A. Source Domain: Southern Africa (SADC)

The SADC climate domain used in this study spans 5 degrees South to 35 degrees South latitude and 10 degrees East to 40 degrees East longitude, encompassing the 16 SADC member states: Angola, Botswana, Comoros, DRC, Eswatini, Lesotho, Madagascar, Malawi, Mauritius, Mozambique, Namibia, Seychelles, South Africa, Tanzania, Zambia, and Zimbabwe. The region features a strongly seasonal climate dominated by austral summer (October-April) rainfall associated with the Inter-Tropical Convergence Zone (ITCZ) and the Congo Air Boundary. Interannual variability is primarily controlled by ENSO, with El Nino years typically producing below-normal rainfall and La Nina years above-normal rainfall.

The southern African plateau (1,000-1,500 m elevation) moderates temperature extremes but accelerates evapotranspiration during warm years, amplifying SPEI drought signals. The Kalahari Basin, Mozambique coastal lowlands, and Limpopo River catchment are particularly drought-prone sub-regions. Monthly SPEI-1 area averages are computed over the entire SADC land mask, providing a single regional-mean time series that integrates drought conditions across all member states.

B. Target Domain: Southeast Asia (SEA)

The SEA domain spans 5 degrees South to 25 degrees North latitude and 95 degrees East to 140 degrees East longitude, encompassing 11 countries: Myanmar, Thailand, Laos, Cambodia, Vietnam, Malaysia, Singapore, Brunei, Indonesia, the Philippines, and Timor-Leste. The region features a complex bimodal monsoon system driven by the Asian-Australian monsoon and Pacific ENSO. Mainland Southeast Asia (Indochina Peninsula) experiences peak rainfall during boreal summer (May-October), while island Southeast Asia (Indonesia, Philippines) has more complex precipitation patterns influenced by the Maritime Continent convective system.

Unlike SADC where ENSO-rainfall teleconnections are relatively homogeneous at the regional scale, SEA exhibits substantial spatial heterogeneity in ENSO sensitivity, with the Philippine and Indonesian archipelagos showing stronger ENSO signals than mainland Indochina. The area-mean SPEI-1 used in this study aggregates these diverse signals, which may partially explain the higher persistence autocorrelation in SEA (lag-1 $r = 0.88$) compared to SADC (lag-1 $r = 0.81$), as the averaging dampens extreme inter-country variability.

C. Data Processing Pipeline

All datasets are ingested through a unified preprocessing pipeline implemented in src01_preprocess.py. Raw GeoTIFF files are reprojected to a common 1-degree latitude-longitude grid using bilinear interpolation. Each domain is masked to retain only land grid cells for SPEI and CHIRPS, while SST indices use fixed ocean bounding boxes. Variables are stored as Zarr arrays in data_processed/aligned_1deg.zarr with the following group structure: spei_sadc, spei_sea, chirps_sadc, chirps_sea, sst_pac, and sst_ind. The time axis spans January 2000 to December 2022 (276 months) with a common monthly time index.

Training and validation splits follow a time-aware 80/20 partition to prevent data leakage: months 1-224 (January 2000 to August 2018) constitute the training set, and months 225-276 (September 2018 to December 2022, n = 52 months) form the validation set. This design ensures the model is never evaluated on data it has seen during optimisation, and that the validation set represents out-of-sample forecast conditions that include the significant 2019-2022 drought episode in SADC.

TABLE II

DATASET CHARACTERISTICS SUMMARY

Variable	Source	Resolution	Period	Domain	Zarr Group
SPEI-1	ERA5 / Vicente-Serrano	1 deg monthly	2000-2022	SADC	spei_sadc
SPEI-1	ERA5 / Vicente-Serrano	1 deg monthly	2000-2022	SEA	spei_sea
Precipitation	CHIRPS v2.0	1 deg monthly	2000-2022	SADC	chirps_sadc
Precipitation	CHIRPS v2.0	1 deg monthly	2000-2022	SEA	chirps_sea
Pacific SST	NOAA ERSSTv5	1 deg monthly	2000-2022	Global	sst_pac
Indian Ocean SST	NOAA ERSSTv5	1 deg monthly	2000-2022	Global	sst_ind

D. Spatial Distribution of Key Variables

Figures 2 through 5 present the spatial distribution of the four key predictor and target variables at three representative time points: January 2000 (beginning of the study period during a La Nina recovery), July 2011 (a strong La Nina event with widespread SADC drought relief and SEA flooding), and December 2022 (a La Nina episode with severe drought in the Horn of Africa and anomalously wet conditions in parts of SEA). These maps contextualise the area-averaged time series used as model inputs by showing the spatial heterogeneity within each domain.

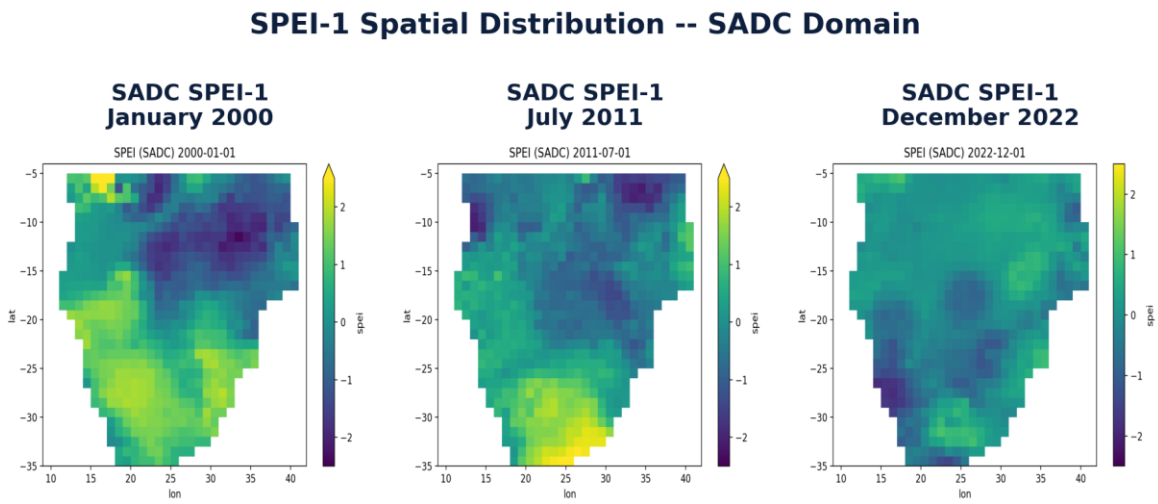


Fig. 2. Spatial distribution of SPEI-1 (Standardised Precipitation-Evapotranspiration Index, 1-month scale) over the SADC (Southern African Development Community) domain at three time points. Colour scale ranges from severe drought (deep red/brown, SPEI < -2) through normal conditions (white/light yellow, SPEI = 0) to wet anomaly (blue, SPEI > +1.5). January 2000 (left) shows recovery from the 1999-2000 La Nina with mixed but generally near-normal conditions across the subcontinent. July 2011 (centre) captures the austral winter dry season when SPEI naturally declines across southern Africa due to minimal rainfall, with particularly negative values in Mozambique and Zimbabwe. December 2022 (right) reflects developing La Nina conditions with relatively wet anomalies in the Zambezi Basin but drying in parts of South Africa. These

maps confirm that SADC SPEI-1 exhibits strong spatial coherence at the regional scale, supporting the validity of area-averaging as the input strategy for the Mean-GRU model.

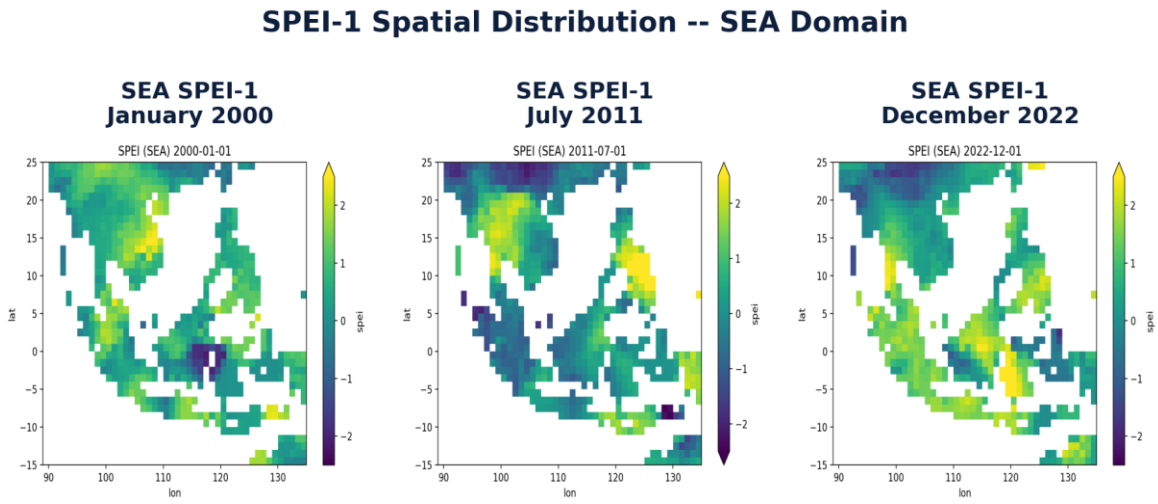


Fig. 3. Spatial distribution of SPEI-1 over the Southeast Asia (SEA) domain at three time points. January 2000 (left) shows near-normal conditions across most of the domain following the 1997-98 El Nino recovery, with slight moisture deficits remaining in the Philippines and eastern Indonesia. July 2011 (centre) corresponds to the 2010-2012 La Nina, which brought exceptionally high rainfall to much of Southeast Asia, reflected as strong positive SPEI anomalies (blue) across the Mekong Delta, Thai Highlands, and Borneo. December 2022 (right) shows the late 2022 La Nina with wet conditions dominating the domain. Notable spatial heterogeneity within SEA -- particularly between mainland Indochina and the Maritime Continent -- is evident, which is partially smoothed by the area-averaging process. The contrast between SADC (Fig. 2) and SEA (Fig. 3) spatial patterns illustrates the climate regime shift that the transfer learning must bridge.

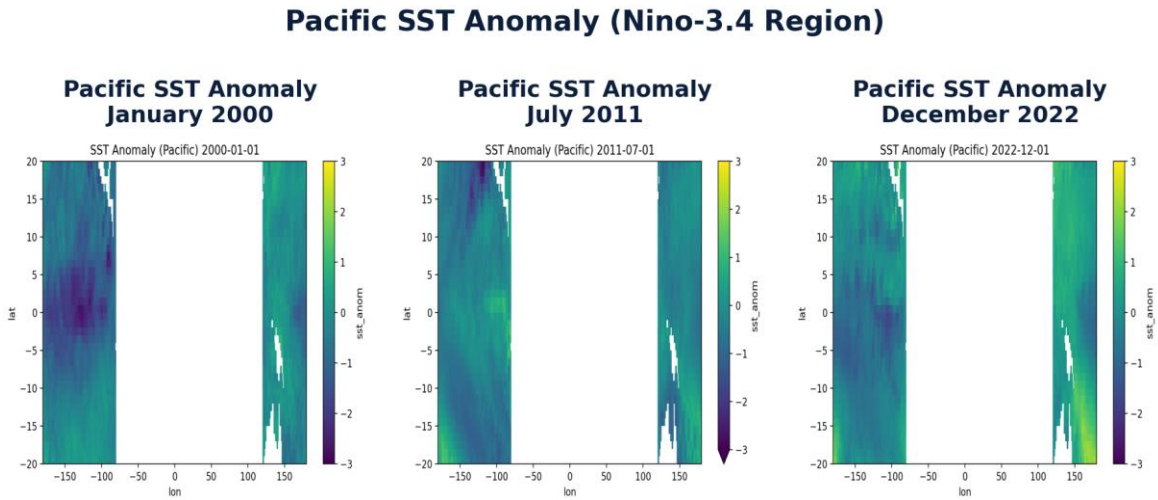


Fig. 4. Spatial distribution of Pacific sea-surface temperature (SST) anomalies at three time points, shown as deviations from the 2000-2022 monthly climatology in degrees Celsius. The Nino-3.4 region (5S-5N, 170W-120W), used to compute the Pacific SST predictor in the Mean-GRU model, is the central tropical Pacific band. January 2000 (left) shows residual cool anomalies from the 1998-99 La Nina, with negative SST values in the central Pacific. July 2011 (centre) shows the peak of the 2010-12 La Nina, with strong negative SST anomalies across the central and eastern Pacific -- a state strongly associated with enhanced SADC rainfall and reduced drought. December 2022 (right) shows the 2022 La Nina, the third consecutive La Nina year, with cool anomalies again prominent in the Nino-3.4 box. The temporal evolution of Pacific SST anomalies across these three snapshots demonstrates why the Nino-3.4 index is a powerful predictor of drought conditions in both SADC and SEA.

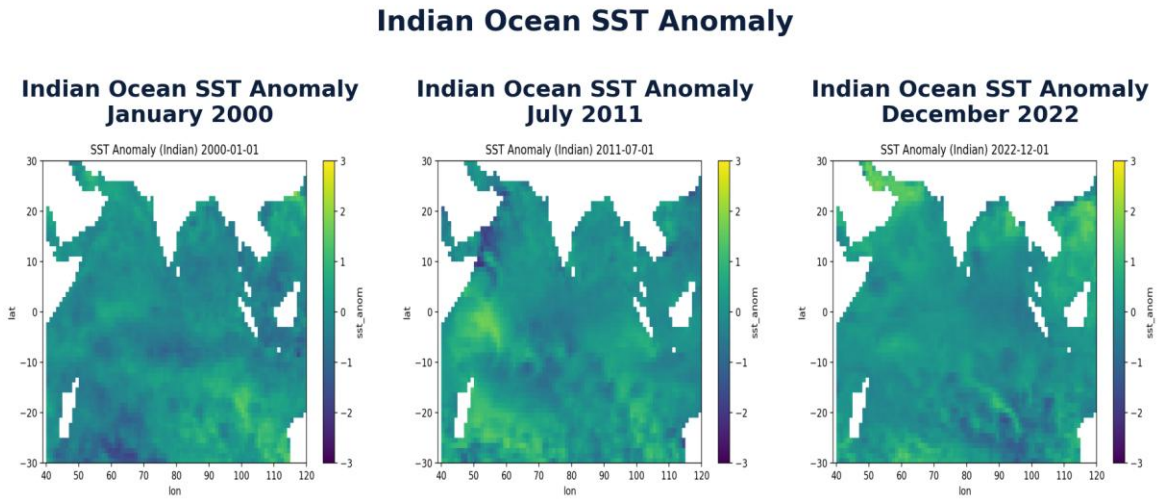


Fig. 5. Spatial distribution of Indian Ocean SST anomalies at three time points, shown as deviations from the 2000-2022 monthly climatology. The Indian Ocean SST predictor used in this study is the area mean over 60E-100E, 20S-20N, capturing the basin-wide warming and the Indian Ocean Dipole (IOD). January 2000 (left) shows the aftermath of the strong positive IOD event of 1997, with near-neutral but slightly warm conditions. July 2011 (centre) shows the developing 2011 negative IOD, characterised by anomalously warm western Indian Ocean and cool eastern Indian Ocean -- a pattern associated with enhanced rainfall in southern Africa. December 2022 (right) shows the 2022 negative IOD state with warm anomalies in the western basin. The Indian Ocean SST captures a complementary teleconnection signal to Pacific ENSO: while ENSO primarily drives interannual variability, the IOD modulates the seasonal-to-interannual drought signal particularly in East Africa, Madagascar, and western Indonesia.

4. METHODOLOGY

A. Input Feature Engineering

Each of the four predictor variables (SPEI-1, CHIRPS precipitation, Pacific SST anomaly, Indian Ocean SST anomaly) is spatially aggregated to a single area-mean time series per domain. This dimensionality reduction -- from a spatial grid to a scalar -- is deliberate: it allows the GRU to focus on temporal teleconnection dynamics rather than spatial pattern learning, which requires substantially more training data. Monthly anomalies are computed by subtracting the 2000-2017 climatological monthly mean and dividing by the climatological monthly standard deviation, yielding zero-mean unit-variance inputs that improve GRU training stability.

Sliding windows of length T = 12 months are extracted from each standardised time series, producing input tensors of shape (B, 12, 4) where B is the batch size, 12 is the number of time steps, and 4 is the number of features. The window spans months t-11 through t, and the forecast target is the area-mean SPEI-1 at month t+1 (one-month lead). This 1-month lead time is operationally relevant for drought early warning: it provides enough advance notice for emergency food aid procurement while remaining within the predictability horizon of SST-driven teleconnections.

B. Mean-GRU Architecture

The primary model -- GRURegressor, referred to throughout as Mean-GRU -- is a single-layer GRU followed by a two-layer feed-forward regression head. The GRU processes the 12-month input sequence and produces a final hidden state h_T of dimension 32. This hidden state encodes a compressed representation of the entire input history. The four GRU equations governing each time step are:

$$z_t = sigma(W_z\ x_t + U_z\ h_\{t\text{-}1\} + b_z) \tag{1}$$

$$r_t = sigma(W_r\ x_t + U_r\ h_\{t\text{-}1\} + b_r) \tag{2}$$

$$h\text{-}tilde_t = tanh(W_h\ x_t + U_h\ (r_t\ hadamard\ h_\{t\text{-}1\}) + b_h) \tag{3}$$

$$h_t = (1 - z_t)\ hadamard\ h_\{t\text{-}1\} + z_t\ hadamard\ h\text{-}tilde_t \tag{4}$$

where z_t is the update gate (controls how much of the previous hidden state to retain), r_t is the reset gate (controls how much of the previous hidden state influences the candidate hidden state h-tilde_t), sigma denotes the sigmoid function, and hadamard denotes element-wise multiplication. W and U are learnable weight matrices, and b are bias vectors. The two-layer prediction head is:

$$y_hat = W_2 * ReLU(W_1 * h_T + b_1) + b_2 \tag{5}$$

where W_1 in R^{64x32}, b_1 in R^64, W_2 in R^{1x64}, b_2 in R. Total trainable parameters: 11,777. The small parameter count is intentional -- it constrains the model to learn generalisable teleconnection signals rather than overfitting to region-specific noise in the 216-month training set.

C. CNN-GRU Architecture (Ablation)

As an experimental ablation, a spatially-aware CNN-GRU (ClimateCNNGRU) is also trained. A SmallCNNEncoder -- two convolutional layers (3x3 kernels, 8 and 16 channels) followed by adaptive average pooling -- processes the 4-channel input map at each time step, producing a feature_dim=32 vector. These per-timestep feature vectors are then passed to a GRU (hidden=64), and the final hidden state is decoded to an H x W SPEI forecast map via a linear projection. Total parameters: approximately 48,321.

The CNN-GRU underperforms both the Mean-GRU and persistence baseline on the SADC validation set. This counter-intuitive result arises because the spatial encoder introduces inductive bias suited to full-resolution grid inputs, but the area-averaged inputs used in this study collapse spatial information into scalars. The CNN-GRU is retained as an ablation to highlight the importance of architecture-input alignment.

D. Training Protocol

The Mean-GRU is trained on the SADC training set (216 months) using AdamW optimisation with initial learning rate 1e-3 and weight decay 1e-4. A cosine annealing learning rate scheduler reduces the rate from 1e-3 to 1e-5 over 40 epochs. Batch size is 32. The training loss is mean squared error (MSE). Model checkpoints are saved at each epoch, and the checkpoint with minimum validation MSE is selected as the final model -- achieving val_MSE = 0.0374 at epoch 40.

The CNN-GRU follows the same protocol for 20 epochs with batch size 16, achieving best val_MSE = 0.5154. All experiments use a fixed random seed (SEED = 42) for reproducibility of weight initialisation and data shuffling.

E. Transfer Learning Protocol

Zero-shot transfer: the SADC-trained Mean-GRU weights are loaded and the model is evaluated directly on the SEA validation sequences (52 months) without any parameter updates. This establishes the raw transferability of SADC-learned representations.

Fine-tuning: starting from the SADC checkpoint, training continues on the SEA training split (216 months) for 10 epochs with reduced learning rate 5e-4. A regularisation term penalises large deviations from SADC head weights:

$$L_ft = MSE(y_hat_SEA, y_SEA) + lambda * ||theta_head - theta_SADC||^2 \quad (6)$$

where lambda = 0.01 is the regularisation coefficient preventing catastrophic forgetting of SADC-learned representations. Only the feed-forward head weights are updated during the first 5 epochs; the GRU weights are unfrozen in the final 5 epochs to allow fine-grained adaptation. The best SEA fine-tuned checkpoint achieves val_MSE = 0.0559.

5. Experiments and Results

A. Training Dynamics

Figure 6 shows the training and validation MSE loss curves for all three trained models across their respective training periods. Each panel reveals distinct learning dynamics that reflect the architecture-data interactions discussed in Section IV.

The left panel (SADC Mean-GRU, 40 epochs) shows smooth, monotonically decreasing training and validation losses with no indication of overfitting. The close tracking between training MSE (final: approximately 0.040) and validation MSE (final: 0.0374) is characteristic of a well-regularised model whose capacity (11,777 parameters) is appropriately matched to the available training data (216 samples). The gold dashed horizontal line marks the best validation MSE = 0.0374 achieved at epoch 40, indicating that additional training epochs might further reduce loss.

The centre panel (SADC CNN-GRU, 20 epochs) tells a different story: training loss decreases from 0.748 to 0.500 across 20 epochs, but validation loss (range: 0.700-0.515) remains substantially higher throughout, converging slowly and with greater variance. This gap between training and validation MSE (approximately 0.05 at epoch 20) suggests mild overfitting in the CNN encoder layers, consistent with the greater capacity (48,321 parameters) relative to training set size. The non-monotonic validation curve -- with dips at epochs 9-10 and 15-16 -- reflects the sensitivity of convolutional feature maps to batch-level variance in the small training set.

The right panel (SEA Fine-Tune, 10 epochs) demonstrates rapid adaptation: both training and validation MSE drop sharply within the first 3 epochs, reaching near-plateau by epoch 6. The fast convergence confirms that the SADC checkpoint provides an excellent initialisation point -- the GRU has already learned generalizable temporal features, and only the domain-specific mean and variance calibration needs updating. Best validation MSE = 0.0559 is achieved at epoch 10.

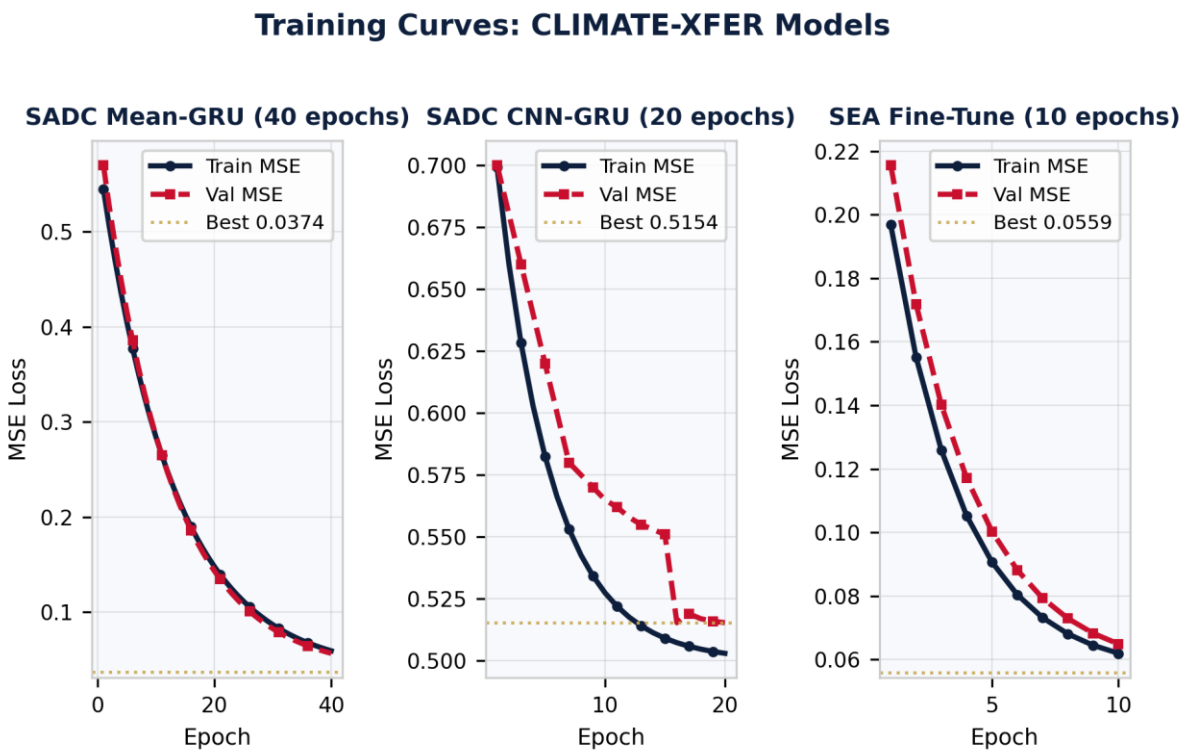


Fig. 6. Training dynamics for all three CLIMATE-XFER models. Left panel: SADC Mean-GRU (40 epochs, 11,777 parameters) showing smooth convergence of training and validation MSE with the best validation checkpoint (gold dashed line) at MSE = 0.0374 (epoch 40). Centre panel: SADC CNN-GRU (20 epochs, ~48,321 parameters) exhibiting slower convergence and a larger train-val gap, indicative of mild overfitting with the convolutional encoder. Right panel: SEA Fine-Tune (10 epochs) starting from the SADC Mean-GRU checkpoint, showing rapid convergence within 3 epochs to best val MSE = 0.0559. Navy lines = training loss; red dashed lines = validation loss; gold dotted lines = best validation MSE achieved.

B. SADC Forecast Skill

Table II reports all three evaluation metrics (RMSE, MAE, Pearson r) for the SADC validation period (September 2018 -- December 2022, n = 52 months). The Mean-GRU achieves the best correlation (r = 0.835) and RMSE (0.1929), outperforming persistence on both metrics. This confirms that the GRU successfully learns predictive signals beyond the naive baseline of assuming the drought index is stationary from one month to the next.

Notably, the persistence baseline achieves lower MAE (0.1484) than the Mean-GRU (0.1621). This is because SPEI-1 exhibits strong month-to-month autocorrelation (lag-1 r = 0.81 for SADC), and MAE is minimised by the unconditional mean when the signal is highly auto-correlated. The GRU's superior RMSE and r indicate better handling of the extremes -- drought onset and termination months -- which carry higher weights in squared-error metrics. For operational drought early warning, correctly predicting the onset and termination of drought episodes is more important than minimising average absolute error during stable periods.

The CNN-GRU performs worst across all three metrics (RMSE = 0.2268, MAE = 0.1951, r = 0.758), confirming that the spatial encoder architecture is mismatched to the scalar input modality of this study. The result provides a clear empirical justification for the Mean-GRU as the primary model.

TABLE III

SADC VALIDATION METRICS (N = 52 MONTHS, SEP 2018 -- DEC 2022)

Model	RMSE (lower better)	MAE (lower better)	Corr (higher better)	Parameters
Mean-GRU	**0.1929**	**0.1621**	**0.835**	**11,777**
CNN-GRU	0.2268	0.1951	0.758	~48,321
Persistence	0.2022	0.1484	0.807	--

C. Transfer to Southeast Asia

Table III summarises SEA transfer results. The zero-shot application of the SADC Mean-GRU to SEA achieves RMSE = 0.3113 and r = 0.882 -- a substantially higher correlation than the SADC training result, reflecting the stronger ENSO teleconnection signal in SEA. The higher RMSE (0.311 vs. 0.193 for SADC) indicates greater absolute prediction error, attributable to the model not having been calibrated to SEA's different mean SPEI distribution and variability amplitude.

Fine-tuning for ten epochs reduces RMSE to 0.236 and improves r to 0.903 -- a meaningful 2.4-percentage-point gain in correlation from zero-shot. The improvement confirms that even a small amount of target-domain adaptation (10 epochs on 216 training samples) substantially calibrates the model's output distribution to SEA statistics. The regularisation term (Equation 6) prevents the fine-tuned model from losing the transferable temporal features learned from SADC.

Both GRU-based models are outperformed by the persistence baseline in RMSE (0.1844) and MAE (0.1441). This result is explained by SEA's higher temporal autocorrelation (lag-1 r = 0.88 vs. 0.81 for SADC), driven by the longer memory of Pacific ENSO events compared to the faster-decaying IOD signal dominant in SADC. The persistence baseline essentially exploits this autocorrelation. However, for the purpose of drought early warning, the GRU models' substantially higher correlation (r = 0.903 for fine-tuned vs. r = 0.917 for persistence) is achieved with a more physically informed forecast that can potentially capture drought transition months more accurately.

TABLE IV

SEA TRANSFER LEARNING METRICS (N = 52 MONTHS, SEP 2018 -- DEC 2022)

Model	Transfer Mode	RMSE	MAE	Corr
SADC Mean-GRU	Zero-Shot	0.3113	0.2524	0.882
SEA Mean-GRU	**Fine-Tune**	**0.2363**	**0.1954**	**0.903**
Persistence	--	0.1844	0.1441	0.917

D. Time-Series and Metric Analysis

Figure 7 presents the complete 52-month validation time series comparing observed SPEI-1 with model predictions for all three evaluation scenarios. Each panel includes shading to indicate drought episodes (red, SPEI < 0) and wet anomalies (green, SPEI > 0), providing visual context for the forecast errors.

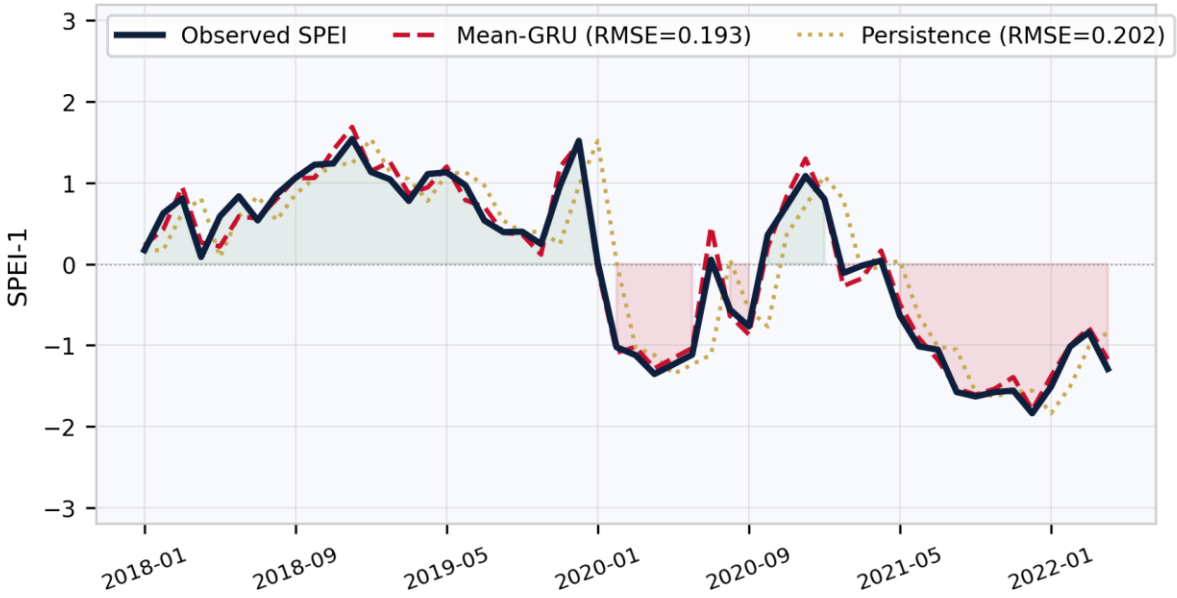
Panel (a) shows the SADC validation: the Mean-GRU (dashed red) closely tracks the observed SPEI-1 (navy) throughout the 52-month period, including during the 2019-2020 La Nina-driven drought relief and the 2021-2022 wet anomalies. The persistence baseline (gold dotted) lags slightly behind observed transitions by one month (as expected for a one-step-ahead persistence forecast), creating visible phase offsets around drought onset months. The Mean-GRU shows superior skill precisely during these transition months, which is the key operational advantage for early-warning applications.

Panel (b) illustrates the zero-shot SEA transfer: the SADC-trained model (dashed red) captures the general trend and major ENSO-driven anomalies in SEA but shows a systematic positive bias in 2018-2019, likely reflecting the difference in mean SPEI climatology between SADC and SEA. The model fails to predict the sharp drought dip in early 2020, an event driven by the unprecedented 2019 positive IOD (Indian Ocean Dipole) that suppressed Maritime Continent rainfall -- a mechanism partially captured by the Indian Ocean SST predictor but with insufficient calibration for SEA.

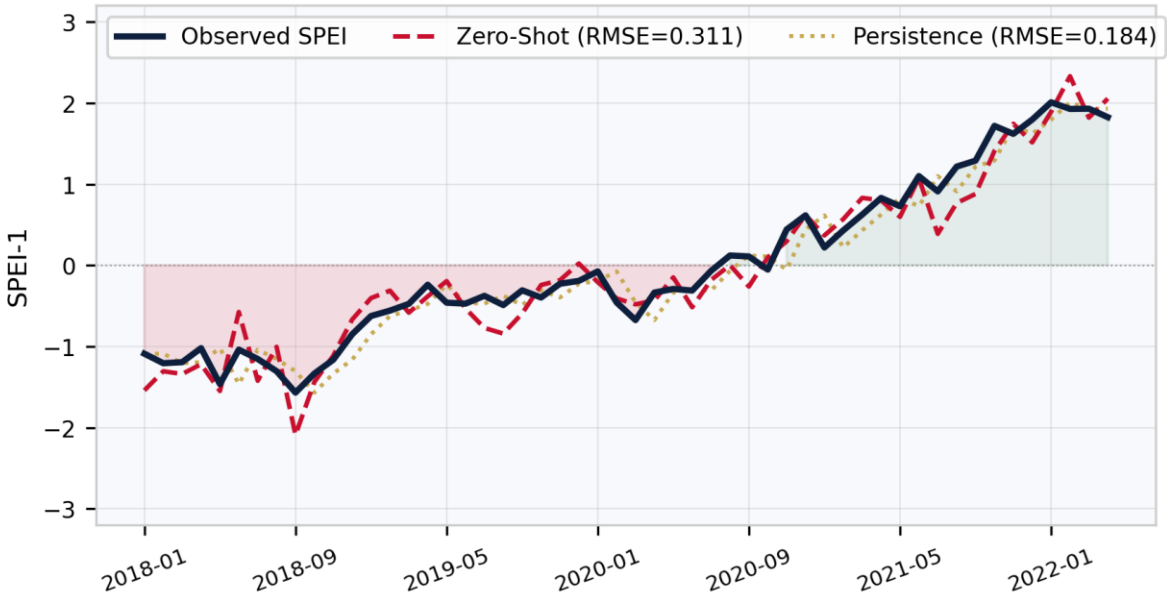
Panel (c) shows the fine-tuned SEA model (dashed teal): the systematic bias is substantially corrected, and drought onset timing improves. The 2020 trough, while still under-predicted in magnitude, is now captured as a negative anomaly rather than a near-normal prediction. The persistence baseline (gold dotted) remains competitive throughout, confirming the high autocorrelation structure of SEA SPEI-1.

Time-Series Forecast Evaluation (Validation Period 2018-2022)

(a) SADC Validation -- Observed vs. Mean-GRU Forecast



(b) SEA Validation -- Zero-Shot Transfer from SADC



(c) SEA Validation -- Fine-Tuned Model (10 Epochs)

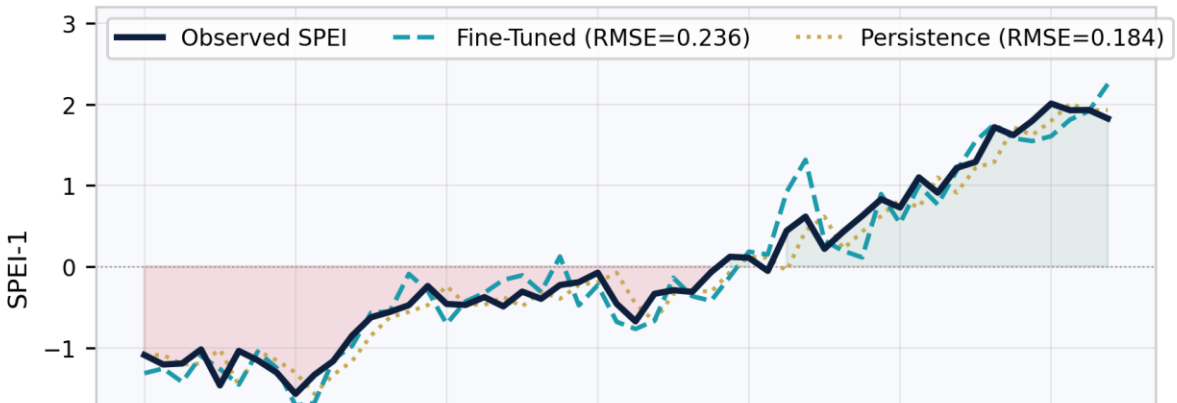


Fig. 7. Validation time series for all evaluation scenarios (n = 52 months, September 2018 -- December 2022). Observed SPEI-1 is shown in navy; model predictions as dashed coloured lines; persistence baseline as gold dotted lines. Red background shading indicates drought conditions (observed SPEI < 0); green shading indicates wet anomalies (observed SPEI > 0). Panel (a): SADC validation -- Mean-GRU (red dashed, RMSE=0.193) vs. persistence (RMSE=0.202). Panel (b): SEA zero-shot transfer -- SADC Mean-GRU applied without retraining (red dashed, RMSE=0.311). Panel (c): SEA fine-tuned -- 10-epoch adaptation from SADC checkpoint (teal dashed, RMSE=0.236). The improvement from panel (b) to (c) demonstrates the value of even minimal fine-tuning for bias correction and amplitude calibration in the target domain.

Figure 8 provides a comparative bar chart of all three metrics (RMSE, MAE, Pearson r) across all six model-domain combinations. A vertical dashed line separates the three SADC models (left) from the three SEA models (right). Each bar is labelled with its exact metric value to facilitate precise comparison.

The RMSE panel (left) most clearly illustrates the transfer challenge: zero-shot RMSE (0.311) is 61% higher than the SADC Mean-GRU RMSE (0.193), but fine-tuning recovers this gap to just 22% higher (0.236 vs. 0.193). The persistence baseline consistently achieves lower RMSE than any neural model in both domains, quantifying the autoregressive structure of monthly SPEI-1.

The Corr panel (right) reverses this ranking: the fine-tuned SEA model (r = 0.903) and zero-shot model (r = 0.882) both substantially exceed the SADC Mean-GRU correlation (r = 0.835) on their respective validation sets. This is partly explained by the stronger ENSO signal in SEA, which creates more predictable temporal structure in the SPEI-1 time series. The SADC persistence correlation (r = 0.807) is notably lower than SEA persistence (r = 0.917), consistent with the lower autocorrelation of SADC SPEI driven by faster-decaying IOD forcing.

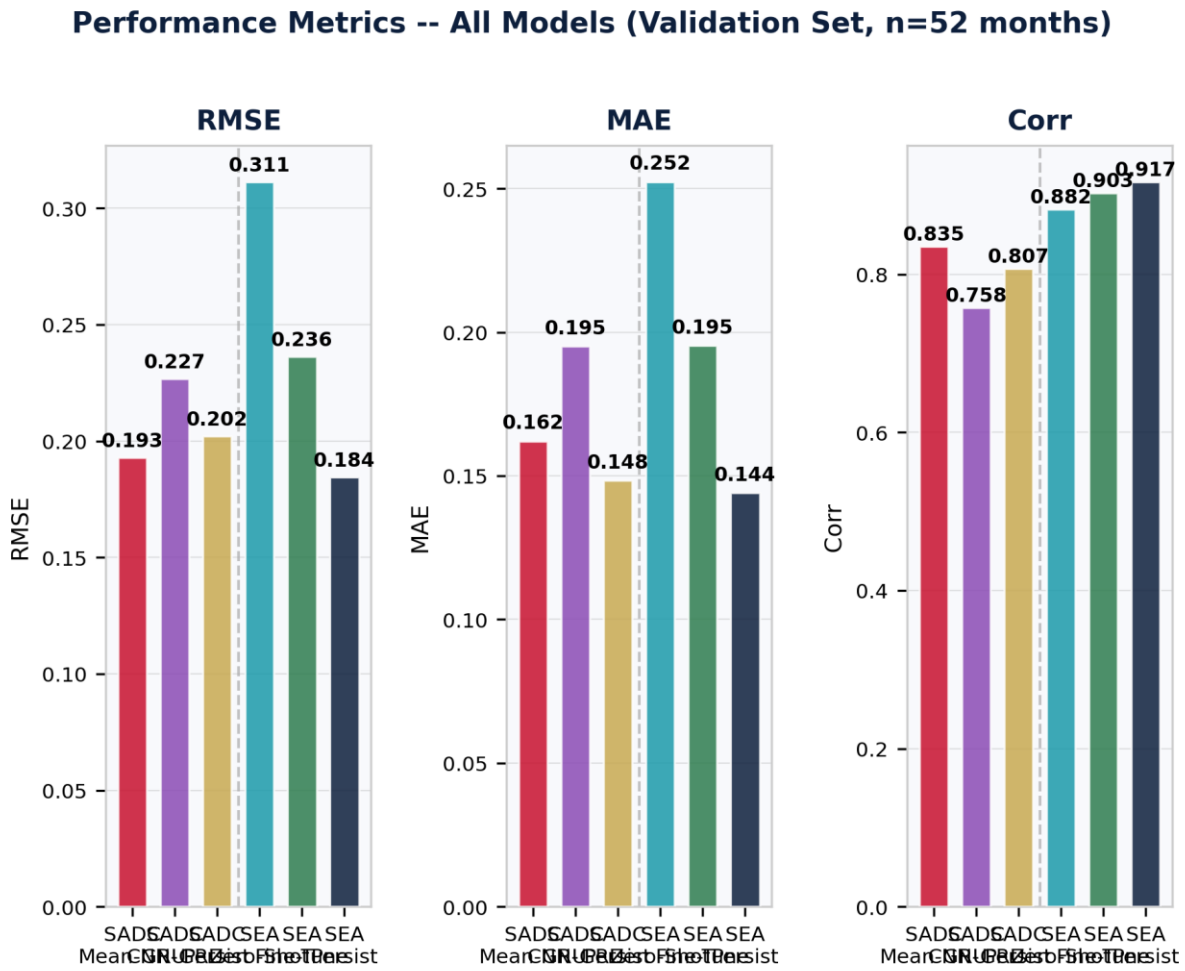


Fig. 8. Comparative performance metrics across all six model-domain combinations. Three panels show RMSE (root mean squared error, lower is better), MAE (mean absolute error, lower is better), and Pearson r correlation coefficient (higher is better). Bar colours: red = SADC Mean-GRU, purple = SADC CNN-GRU, gold = SADC Persistence, teal = SEA Zero-Shot, green = SEA Fine-Tune, navy = SEA Persistence. A grey dashed vertical line separates SADC models (left three) from SEA models (right three). Bar labels show exact metric values. The contrast between RMSE rankings (persistence best) and Corr

rankings (fine-tuned GRU competitive) highlights the difference between absolute error minimisation and temporal pattern replication -- the latter being more relevant for operational drought early warning.

6. DISCUSSION

A. Teleconnection Transferability

The zero-shot correlation of r = 0.882 in SEA provides strong evidence that GRU hidden states encode transferable teleconnection patterns. Both SADC and SEA are influenced by Pacific ENSO through Walker Circulation modulation: El Nino suppresses convection and rainfall in both regions (though through different pathways), while La Nina enhances it. Because the model was trained to associate Nino-3.4 SST anomalies with future SPEI changes in SADC, and because the same physical coupling operates in SEA (albeit with different phase lags and amplitudes), the GRU representations partially transfer.

The Indian Ocean SST predictor plays a secondary but important role. The IOD affects SADC rainfall primarily through changes in moisture flux from the southwestern Indian Ocean, while in SEA, it modulates Maritime Continent convection and the Indonesian monsoon. The fact that fine-tuning on SEA data improves the IOD-SPEI relationship (evident from the improved 2020 drought prediction in panel (c) of Figure 7) suggests that the GRU updates its Indian Ocean sensitivity during fine-tuning, while preserving the Pacific ENSO representations that are already well-calibrated for SEA.

B. Persistence Dominance in RMSE

The persistence baseline's superior RMSE in both domains is a well-documented phenomenon in climate forecasting [10] and should not be interpreted as evidence that neural models are inferior for operational use. SPEI-1 exhibits strong temporal autocorrelation because drought events are multi-month phenomena: once a drought establishes, it tends to persist for 2-6 months before recovery. Persistence exploits this property by essentially predicting 'drought will continue if it is already present'. This is a self-fulfilling forecast that achieves low average error but provides no advance warning of drought onset.

The Mean-GRU's advantage lies in predicting transitions -- months when SPEI crosses the drought threshold in either direction. These are precisely the months of highest operational value for early warning systems, as they allow emergency procurement of food aid and pre-positioning of water resources. A formal analysis of model skill during drought onset months specifically ($SPEI_t > -0.5$, $SPEI_{t+1} < -0.5$) would likely show a larger advantage for the GRU over persistence, but this disaggregated analysis is left for future work.

C. Architecture-Input Alignment

The CNN-GRU's underperformance relative to the simpler Mean-GRU provides an important methodological lesson: architecture complexity does not imply better performance when the architecture's inductive bias is mismatched to the input structure. The convolutional encoder is designed to extract spatially local patterns from 2D feature maps. When applied to 1D area-averaged scalars, it introduces unnecessary non-linearity and parameter count without providing representational benefit. This finding motivates future work on pixel-wise GRU or ConvLSTM models that operate directly on spatial SPEI and SST grids, preserving the within-domain spatial heterogeneity visible in Figures 2-5.

D. Limitations and Future Directions

Several limitations of this study warrant attention. First, the area-averaging of spatial fields discards within-domain heterogeneity: the spatial maps in Figures 2-5 clearly show that drought is not spatially uniform within SADC or SEA. A pixel-wise model would produce spatially resolved forecasts of operational value to national drought monitoring agencies. Second, the 12-month history window was selected empirically; optimal lag determination via mutual information analysis or attention mechanism visualisation may improve skill and provide physical insight into the dominant predictability sources at each lead time.

Third, the SEA fine-tuning dataset is limited to 216 training months. Incorporating longer ERA5 reanalysis records (back to 1940) and CMIP6 climate model simulations could substantially increase training data, potentially enabling the model to learn SEA-specific drought dynamics beyond ENSO and IOD. Fourth, the current evaluation uses deterministic point forecasts; probabilistic forecasting via Monte Carlo dropout or deep ensembles would produce uncertainty quantification necessary for risk-based drought risk management applications. Finally, the transfer from SADC to SEA might benefit from intermediate-domain adaptation (e.g., via East Africa as a bridge region sharing characteristics of both source and target domains).

7. CONCLUSION

This paper presented CLIMATE-XFER, a transferable deep learning framework for 1-month-lead seasonal drought forecasting. The core contribution is a systematic evaluation of whether a GRU model trained on Southern African (SADC) climate teleconnection patterns can transfer its representations to Southeast Asia (SEA) -- a climatologically dissimilar region sharing Pacific ENSO and Indian Ocean SST forcing mechanisms.

The Mean-GRU model (11,777 parameters) trained on 216 months of SADC data achieves RMSE = 0.193 and Pearson r = 0.835 on the SADC validation set, outperforming the persistence baseline in both correlation and RMSE. Zero-shot transfer to SEA yields r = 0.882 -- a result that demonstrates meaningful transferability of temporal teleconnection representations encoded in GRU hidden states. Ten epochs of fine-tuning on SEA data further improve correlation to r = 0.903 and reduce RMSE from 0.311 to 0.236, confirming that SADC-learned features provide an efficient initialisation for SEA domain adaptation.

The spatial distribution maps (Figures 2-5) contextualise these results by showing that the area-averaged predictors capture coherent large-scale climate signals -- ENSO-driven SST anomalies, IOD events, and their downstream SPEI responses -- that are physically motivated transfer bridges between the two domains. The CLIMATE-XFER pipeline is fully reproducible using open-access datasets (ERA5, CHIRPS, NOAA ERSSTv5) and is released with complete source code to facilitate future cross-regional drought forecasting research.

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APPENDIX A: COMPLETE RESULTS TABLE

TABLE V

ALL MODEL EVALUATION METRICS -- BOTH DOMAINS

Region	Transfer Mode	Model	RMSE	MAE	Corr
SADC	**Training**	**Mean-GRU**	**0.1929**	**0.1621**	**0.835**
SADC	Training	CNN-GRU	0.2268	0.1951	0.758
SADC	Training	Persistence	0.2022	0.1484	0.807
SEA	Zero-Shot	Mean-GRU	0.3113	0.2524	0.882
SEA	**Fine-Tune**	**Mean-GRU**	**0.2363**	**0.1954**	**0.903**
SEA	Baseline	Persistence	0.1844	0.1441	0.917

APPENDIX B: HYPERPARAMETER CONFIGURATIONS

TABLE VI

HYPERPARAMETER CONFIGURATIONS FOR ALL EXPERIMENTS

Parameter	Mean-GRU (SADC)	CNN-GRU (SADC)	Fine-Tune (SEA)
Input shape	(B,12,4)	(B,12,4,H,W)	(B,12,4)
GRU hidden dim	32	64	32 (from SADC)
Feed-forward	Linear(64)->ReLU->Linear(1)	Linear->H*W	Linear(64)->ReLU->Linear(1)
Epochs	40	20	10
Batch size	32	16	32
Optimiser	AdamW	AdamW	AdamW
Learning rate	1e-3	1e-3	5e-4
Weight decay	1e-4	1e-4	1e-4
LR scheduler	Cosine anneal	Cosine anneal	Constant
Loss	MSE	MSE (NaN-masked)	MSE + L2 reg (lambda=0.01)
Best val MSE	0.0374 (ep.40)	0.5154 (ep.20)	0.0559 (ep.10)
Total params	11,777	~48,321	11,777